

1. A charge of 120 C passes through a conductor in 2 minutes. Calculate the current flowing through the conductor.
2. A current of 2.5 A flows through a wire for 4 minutes. How much charge passes through any cross-section of the wire?
3. A current of 3.2 A flows through a conductor for 10 s. Calculate the number of electrons passing through a cross-section of the conductor. ($e = 1.6 \times 10^{-19}$ C)
4. The current through a conductor varies with time according to $I = (4 + 2t)$ A, where t is in seconds. Find the total charge passing through the conductor between $t = 0$ and $t = 5$ s.
5. 600 J of work is required to move a charge of 50 C between two points in a conductor. Find the potential difference between the two points.
6. A charge of 5 C moves through a potential difference of 12 V. Calculate the work done by the electric field.
7. A potential difference of 24 V is applied across a resistor and a current of 3 A flows through it. Find its resistance.
8. A conductor has a resistance of 20 Ω . What current will flow through it when a potential difference of 100 V is applied?
9. A copper wire has a cross-sectional area of 1 mm². It carries a current of 3.2 A. If the number density of free electrons is 8×10^{28} m⁻³, calculate the drift velocity of electrons. Use $I = neAv_d$.
10. A wire carries a steady current I . If its cross-sectional area is reduced to half while the current remains unchanged, how does its drift velocity change, assuming the electron number density remains constant?
11. The drift velocity of electrons in a conductor is 0.5 mm s⁻¹. If the number density of electrons is 5×10^{28} m⁻³ and the cross-sectional area is 2 mm², calculate the current.
12. In a metallic conductor, the electrons move randomly with very large speeds even when no current flows. When a potential difference is applied, they acquire a small drift velocity. Explain why the current is determined by the drift velocity rather than by the random velocity of electrons.
13. A wire of length 2 m and cross-sectional area 0.5 mm² has resistivity $\rho = 1.6 \times 10^{-8}$ Ω m. Calculate its resistance.
14. A wire has a resistance of 10 Ω , length 5 m, and cross-sectional area 2×10^{-6} m². Calculate its resistivity.

15. A wire of resistance R is stretched uniformly so that its length becomes twice its original length. Assuming its volume remains constant, find its new resistance in terms of R .

16. Two wires are made of the same material. Wire A has length L and radius r , while wire B has length $2L$ and radius $2r$. Find the ratio of their resistances, R_A/R_B .

17. The resistance of a metallic conductor is $10\ \Omega$ at $20\ ^\circ\text{C}$. Its temperature coefficient of resistance is $4 \times 10^{-3}\ ^\circ\text{C}^{-1}$. Calculate its resistance at $70\ ^\circ\text{C}$, assuming the temperature coefficient remains constant.

18. A student claims that if the length of a wire is doubled, its resistivity also doubles. Is this statement correct? Explain the distinction between resistance and resistivity, and state which physical quantities each depends upon.

19. Starting from $I = neAv_d$ and $v_d = eE\tau/m$, derive the relation $J = \sigma E$, and obtain an expression for the conductivity σ in terms of n , e , m and τ .

20. A metallic wire of length L and cross-sectional area A is connected across a potential difference V . The number density of free electrons is n , and the relaxation time is τ . (a) Derive an expression for the drift velocity of electrons. (b) Derive an expression for the current through the wire. (c) Show that the resistance of the wire can be written as $R = mL/(ne^2\tau A)$. (d) Hence obtain the expression for the resistivity of the material.

21. A current of $0.5\ \text{A}$ flows through a conductor. How many electrons pass through a cross-section of the conductor in 1 minute? Take $e = 1.6 \times 10^{-19}\ \text{C}$.

22. The potential difference between two points is $15\ \text{V}$. How much work is required to move a charge of $8\ \text{C}$ from one point to the other?

23. A wire carries a current of $4\ \text{A}$ when a potential difference of $20\ \text{V}$ is applied across it. If the potential difference is increased to $50\ \text{V}$ and the resistance remains unchanged, find the new current.

24. A uniform wire has resistance $12\ \Omega$. It is cut into three equal lengths. Find the resistance of each piece.

25. A wire of resistance $10\ \Omega$ is cut into two equal parts and the two parts are connected in parallel. Find the equivalent resistance.

26. A conductor has length $4\ \text{m}$, cross-sectional area $1.5 \times 10^{-6}\ \text{m}^2$, and resistance $8\ \Omega$. Calculate the resistivity of the material.

27. Two wires of the same material have equal lengths, but the radius of one wire is twice that of the other. Find the ratio of their resistances.

28. A wire has resistance R at room temperature. If its length is increased by 20% without changing its volume, find its new resistance in terms of R .

29. The number density of free electrons in a metal is $6 \times 10^{28} \text{ m}^{-3}$. If the drift velocity is $2 \times 10^{-4} \text{ m s}^{-1}$ and the cross-sectional area of the wire is 1 mm^2 , calculate the current.

Take $e = 1.6 \times 10^{-19} \text{ C}$.

30. A potential difference V is applied across a metallic wire. If the number density of free electrons is n , cross-sectional area is A , and drift velocity is v_d , derive the expression for current and explain how the current changes if A is doubled while V remains unchanged.